

Banach Spaces

Functional Analysis

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Definition

Suppose M is any set. A **metric on M** is a function $d : M \times M \rightarrow \mathbb{R}$ satisfying the following properties for all $x, y, z \in M$:

- i) **(Symmetry)** $d(x, y) = d(y, x)$.
- ii) **(Positive-definite)** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
- iii) **(Triangle inequality)** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (M, d) is called a **metric space**. When the metric is understood from the context, one often just says “ M is a metric space”.

Definition

For any $x \in M$ and $r > 0$, the **(open) ball of radius r around x** is the set

$$B_r(x) = \{y \in M : d(y, x) < r\},$$

and the **closed ball of radius r around x** is

$$\overline{B}_r(x) = \{y \in M : d(y, x) \leq r\}.$$

Definition

A subset $A \subseteq M$ of a metric space M is said to be an **open subset of M** if it contains an open ball around each of its points, and it is said to be a **closed subset of M** if $M \setminus A$ is open.

Definition

A **sequence** in M is any map $X : \mathbb{N} \rightarrow M$. A sequence (x_n) of points in a metric space M is said to **converge** to $x \in M$ if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < \varepsilon \quad \text{for } n \geq N.$$

In this case, we call x the **limit of the sequence** and write

$$x_n \rightarrow x \quad \text{or} \quad \lim(x_n) = x.$$

A sequence is **divergent** if it does not converge to any point. A sequence (x_n) in a metric space is said to be **bounded** if its image $\{x_n\}_{n \in \mathbb{N}}$ is a bounded subset of M .

Remark

We will usually denote a generic metric space by symbols such as $\mathcal{V}, \mathcal{W}, \mathcal{U}$ and its generic elements by symbols such as f, g, h because many of the important metric spaces in analysis are sets of functions that form linear spaces.

Theorem 1.1

Let (V, d) be a metric space and $E \subseteq V$. Then

- i) $\overline{E} = \{g \in V : B_\varepsilon(g) \cap E \neq \emptyset \text{ for all } \varepsilon > 0\};$
- ii) $\overline{E} = \{g \in V : \text{there exists a sequence in } E \text{ convergent to } g\};$
- iii) $\overline{E}$ is a closed subset of V ;
- iv) E is closed if and only if $\overline{E} = E$;
- v) E is closed if and only if E contains the limit of every convergent sequence of elements of E .

Definition

If (M_1, d_1) and (M_2, d_2) are metric spaces and f is a point in M_1 , a map $T : M_1 \rightarrow M_2$ is said to be **continuous at f** if for any $\varepsilon > 0$ there exists $\delta > 0$ such that $d_1(f, g) < \delta$ implies $d_2(T(f), T(g)) < \varepsilon$ for all $g \in M_1$; and T is said to be **continuous** if it is continuous at each point in M_1 .

Theorem 1.2

Suppose $\mathcal{V}$ and $\mathcal{W}$ are metric spaces equipped with their metric topologies $\mathcal{T}_{\mathcal{V}}, \mathcal{T}_{\mathcal{W}}$ and let $T \in \text{Hom}(\mathcal{V}, \mathcal{W})$. Then the following are equivalent

- i) *T is continuous.*
- ii) *If $\lim_{\mathcal{V}} f_n = f$, then $\lim_{\mathcal{W}} Tf_n = Tf$.*
- iii) *$T^{-1}(\mathcal{T}_{\mathcal{W}}) \subseteq \mathcal{T}_{\mathcal{V}}$.*
- iv) *$T^{-1}(\mathcal{T}_{\mathcal{W}}^{\sim}) \subseteq \mathcal{T}_{\mathcal{V}}^{\sim}$.*

Here, $\mathcal{T}^{\sim}$ is the collection of closed subsets.

Definition

A sequence (f_n) in a metric space is said to be **Cauchy** (or a **Cauchy sequence**) if for every $\varepsilon > 0$, there exists $H \in \mathbb{N}$ such that

$$d(f_n, f_m) < \varepsilon \quad \text{for } n, m \geq H.$$

A metric space in which every Cauchy sequence converges to a value admitted in the space is said to be **complete**.

Theorem 1.3

A complete subset of a metric space is closed and a closed subset of a complete metric space is complete.

Definition

A **norm** on a linear space $\mathcal{V}$ is a functional $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{R}$ satisfying the following properties:

- i) **(Positive definiteness)** $\|v\| \geq 0$ for all $v \in \mathcal{V}$ and $\|v\| = 0$ if and only if $v = 0$.
- ii) **(Homogeneity)** $\|\alpha v\| = |\alpha| \|v\|$ for any complex number α .
- iii) **(Triangle Inequality)** $\|u + v\| \leq \|u\| + \|v\|$.

A linear space on which a norm is defined is called a **normed linear space**. If a functional satisfies the previous properties but with

- i') **(Nonnegativeness)** $\|v\| \geq 0$ for all $v \in \mathcal{V}$
- replacing i), then it is called a **pseudonorm**.

Theorem 1.4 (Norms Induce Metrics)

Let $(\mathcal{V}, \|\cdot\|)$ be a normed linear space. Then the map $d : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ defined by

$$d(f, g) = \|f - g\|$$

*is a metric on $\mathcal{V}$ called the **induced metric**.*

Definition

Let $\mathcal{V}$ be a normed linear space and let d be the induced metric. Then a sequence (f_n) in $\mathcal{V}$ is said to be **convergent** [respectively, **Cauchy**] if it is convergent [respectively, Cauchy] in the metric space $(\mathcal{V}, d)$.

Definition

Let $\mathcal{V}$ be a normed space with respect to two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then we say that $\|\cdot\|_a$ and $\|\cdot\|_b$ are **equivalent norms** if they induce the same topologies on $\mathcal{V}$.

Theorem 1.5

Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be two norms on $\mathcal{V}$. Then the following statements are equivalent:

- i) $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent norms*
- ii) $\|\cdot\|_a$ and $\|\cdot\|_b$ define the same convergence criterion.*
- iii) There exist constants $C_1, C_2 > 0$ such that*

$$C_1\|f\|_a \leq \|f\|_b \leq C_2\|f\|_a \quad \text{for all } f \in \mathcal{V}.$$

Definition

*A complete normed linear space is called a **Banach space**.*

Definition

If (f_n) is a sequence in a normed linear space $\mathcal{V}$, then the **formal infinite series** (or simply the **series**) **generated by** (f_n) is the sequence $S := (s_k)$ defined by

$$s_1 := f_1$$

$$s_2 := s_1(z) + f_2 \quad (= f_1 + f_2)$$

$$\vdots$$

$$s_k := s_{k-1}(z) + f_k \quad (= f_1 + f_2 + \cdots + f_k)$$

$$\vdots$$

The numbers f_n are called the **terms** of the series and the numbers s_k are called the **partial sums** of this series. If $\lim S$ exists, we say that this series is **convergent** and call this limit the **sum** or the **value** of this series. If this limit does not exist, we say that the series S is **divergent**. It is convenient to use symbols such as

$$\sum(f_n) \quad \text{or} \quad \sum f_n \quad \text{or} \quad \sum_{n=1}^{\infty} f_n$$

to denote both the formal infinite series S generated by the sequence (f_n) and also to denote the value $\lim S$, in case this limit exists.

Definition

Let (f_n) be a sequence in $\mathcal{V}$. We say that the series $\sum f_n$ is **absolutely convergent** if the series $\sum \|f_n\|$ is convergent in $\mathbb{R}$. A series is said to be **conditionally (or nonabsolutely) convergent** if it is convergent, but it is not absolutely convergent.

Theorem 1.6

Let $\mathcal{V}$ be a normed linear space. Then $\mathcal{V}$ is a Banach space if and only if absolute convergence implies nonabsolutely convergence for any series generated by sequences in $\mathcal{V}$.

Definition

Let $\mathcal{V}, \mathcal{W}$ be normed spaces. We define the **norm** of an element $T \in \text{Hom}(\mathcal{V}, \mathcal{W})$ to be the number

$$\|T\| = \sup \left\{ \frac{\|Tf\|}{\|f\|} : f \in \mathcal{V}, f \neq 0 \right\}.$$

An element $T \in \text{Hom}(\mathcal{V}, \mathcal{W})$ is said to be **bounded** if $\|T\| < \infty$ and the family of all such bounded maps is denoted by $\text{Hom}_b(\mathcal{V}, \mathcal{W})$ or $\mathcal{B}(\mathcal{V}, \mathcal{W})$.

Theorem 1.7

$$\begin{aligned}\|T\| &= \sup \{ \|Tf\| : f \in \mathcal{V} \text{ and } \|f\| \leq 1 \} \\ &= \sup \{ \|Tf\| : f \in \mathcal{V} \text{ and } \|f\| < 1 \} = \sup \{ \|Tf\| : f \in \mathcal{V} \text{ and } \|f\| = 1 \} \\ &= \inf \{ c \in [0, \infty) : \|Tf\| \leq c\|f\| \text{ for all } f \in \mathcal{V} \}.\end{aligned}$$

Theorem 1.8

Suppose $\mathcal{V}$ and $\mathcal{W}$ are normed spaces. Then $\|\cdot\|$ is a norm on $\text{Hom}_b(\mathcal{V}, \mathcal{W})$.

Theorem 1.9

Suppose $\mathcal{V}$ is a normed space and $\mathcal{W}$ is a Banach space. Then $\text{Hom}_b(\mathcal{V}, \mathcal{W})$ is a Banach space.

Theorem 1.10

For any $\mathcal{V}, \mathcal{W}$ normed spaces, an element $T \in \text{Hom}(\mathcal{V}, \mathcal{W})$ is continuous if and only if it is bounded.

Definition

Let $\mathcal{V}$ be a linear space over $\mathbb{K}$. A **linear functional** on $\mathcal{V}$ is an element of $\text{Hom}(\mathcal{V}, \mathbb{K})$.

Theorem 1.11

Suppose $\mathcal{V}$ is a normed space and let φ be a nonzero linear functional. Then the following are equivalent:

- i) φ is a bounded linear functional.
- ii) φ is a continuous linear functional.
- iii) $\ker \varphi$ is closed in $\mathcal{V}$.
- iv) $\overline{\ker \varphi} \neq \mathcal{V}$.

Definition

Suppose $\{e_\alpha\}_{\alpha \in I}$ is a family in a linear space $\mathcal{V}$. Then

- a) $\{e_\alpha\}_{\alpha \in I}$ is called **linearly independent** if there does not exist a finite nonempty subset $J \subseteq I$ and a family $\{\alpha_j\}_{j \in J}$ in $\mathbb{K} \setminus \{0\}$ such that

$$\sum_{j \in J} \alpha_j e_j = 0.$$

- b) The **span** of $\{e_\alpha\}_{\alpha \in I}$, denoted by $\text{Span}_{\mathbb{K}}\{e_\alpha\}_{\alpha \in I}$ is the set of all sums of the form

$$\sum_{j \in J} \alpha_j e_j,$$

where J is a finite subset of I and $\{\alpha_j\}_{j \in J}$ is a family in $\mathbb{K}$.

- c) A linear space $\mathcal{V}$ over $\mathbb{K}$ is called **finite-dimensional** if there exists a finite set I and a family $\{e_\alpha\}_{\alpha \in I}$ in $\mathcal{V}$ such that $\text{Span}_{\mathbb{K}}\{e_\alpha\}_{\alpha \in I} = \mathcal{V}$.
- d) A linear space is called **infinite-dimensional** if it is not finite-dimensional
- e) A family in $\mathcal{V}$ is called a **Hamel basis** of $\mathcal{V}$ if it is linearly independent and its span equals $\mathcal{V}$.

Definition

*Suppose $\mathcal{A}$ is a collection of subsets of $\mathcal{V}$. A set $\Gamma \in \mathcal{A}$ is called a **maximal** element of $\mathcal{A}$ if there does not exist $\Gamma' \in \mathcal{A}$ with $\Gamma \subset \Gamma'$.*

Theorem 1.12

Suppose $\mathcal{V}$ is a linear space. Then a subset of $\mathcal{V}$ is a basis of $\mathcal{V}$ if and only if it is a maximal element of the collection of linearly independent subsets of $\mathcal{V}$.

Definition

A collection $\mathcal{C}$ of subsets of a set $\mathcal{V}$ is called a **chain** if $\Omega, \Gamma \in \mathcal{C}$ implies $\Omega \subseteq \Gamma$ or $\Gamma \subseteq \Omega$.

Lemma 1.13 (Zorn's Lemma)

Suppose $\mathcal{V}$ is a set and $\mathcal{A}$ is a collection of subsets of $\mathcal{V}$ with the property that the union of all the sets in $\mathcal{C}$ is in $\mathcal{A}$ for every chain $\mathcal{C} \subseteq \mathcal{A}$. Then $\mathcal{A}$ contains a maximal element.

Theorem 1.14

Every linear space has a Hamel basis.

Theorem 1.15

If $\mathcal{V}$ is a finite-dimensional linear space, then all norms on $\mathcal{V}$ are equivalent.

Theorem 1.16

Let $\mathcal{V}$ be a finite-dimensional normed space and let $\mathcal{W}$ be any normed linear space. Then every element of $\text{Hom}(\mathcal{V}, \mathcal{W})$ is continuous.

Theorem 1.17

Every infinite-dimensional normed space has a discontinuous linear functional.

Lemma 1.18 (Extension Lemma)

Suppose $\mathcal{V}$ is a real normed space, $\mathcal{U}$ is a subspace of $\mathcal{V}$, and $\psi : \mathcal{U} \rightarrow \mathbb{R}$ is a bounded linear functional. Suppose $h \in \mathcal{V}/\mathcal{U}$. Then ψ can be extended to a bounded linear functional $\varphi : \mathcal{U} + \mathbb{R}h \rightarrow \mathbb{R}$ such that $\|\varphi\| = \|\psi\|$.

Definition

Suppose $T : \mathcal{V} \rightarrow \mathcal{W}$ is a function from a set $\mathcal{V}$ to a set $\mathcal{W}$. Then the **graph** of T , denoted by $\text{graph}(T)$, is the subset of $\mathcal{V} \times \mathcal{W}$ defined by

$$\text{graph}(T) = \{(f, T(f)) : f \in \mathcal{V}\}.$$

Theorem 1.19

Suppose $\mathcal{V}$ and $\mathcal{W}$ are normed spaces and $T : \mathcal{V} \rightarrow \mathcal{W}$ is a function.

- i) T is an element of $\text{Hom}(\mathcal{V}, \mathcal{W})$ if and only if $\text{graph}(T)$ is a subspace of $\mathcal{V} \times \mathcal{W}$.*
- ii) Suppose $\mathcal{U} \subseteq \mathcal{V}$ and $S : \mathcal{U} \rightarrow \mathcal{W}$ is a function. Then T is an extension of S if and only if $\text{graph}(S) \subseteq \text{graph}(T)$.*
- iii) If $T \in \text{Hom}(\mathcal{V}, \mathcal{W})$ and $c \in [0, \infty)$, then $\|T\| \leq c$ if and only if $\|g\| \leq c\|f\|$ for all $(f, g) \in \text{graph}(T)$.*

Theorem 1.20 (Hahn-Banach Theorem)

Suppose $\mathcal{V}$ is a normed space, $\mathcal{U}$ is a subspace of $\mathcal{V}$, and $\psi : \mathcal{U} \rightarrow \mathbb{K}$ is a bounded linear functional. Then ψ can be extended to a bounded linear functional on $\mathcal{V}$ whose norm equals $\|\psi\|$.

Definition

Suppose $\mathcal{V}$ is a normed space. Then the **dual space** $\mathcal{V}^*$ of V is the normed space of all bounded linear functionals on $\mathcal{V}$.

Corollary

If $\mathcal{V}$ is any normed space, then $\mathcal{V}^$ is a Banach space.*

Theorem 1.21

Suppose $\mathcal{V}$ is a normed space and let $f \in \mathcal{V} \setminus \{0\}$. Then there exists $\varphi \in \mathcal{V}^$ such that $\|\varphi\| = 1$ and $\|f\| = \varphi(f)$.*

Theorem 1.22

Suppose $\mathcal{U}$ is a subspace of a normed space and $h \in \mathcal{V}$. Then $h \in \overline{\mathcal{U}}$ if and only if $\varphi(h) = 0$ for every $\varphi \in \mathcal{V}^$ such that $\varphi|_{\mathcal{U}} = 0$.*